A Minor Project Synopsis

On

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Face Detection And Recognition System

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FACE DETECTION AND RECOGNIZATION SYSTEM

Introduction

In today's digital world identifying people accurately is a very big challenge for security and automation. Biometrics which uses physical features like fingerprints and facial features have become the most reliable way to verify identity of a person. Unlike passwords or ID cards, your face data cannot be lost, stolen, or forgotten easily.

It is important to understand that this project performs two distinct tasks:

- Face Detection: The system first scans a video frame to find *where* a human face is located. It distinguishes a face from the background (like a wall or a chair).
- Face Recognition: Once a face is found, the system analyses the unique features like distance between eyes, shape of the jaw, texture of the skin to decide who that person is by comparing them to a saved database in the system.

By using OpenCV which is an open source computer vision library and Python, we can build a system that works in real-time to recognize facial features. This helps the computer to processes the video so fast that it identifies the person almost instantly when they step in front of the camera.

Traditional systems, such as manual attendance registers or barcode scanners, are slow, outdated and require physical contact. Especially in a post-pandemic world, contactless systems are preferred. This project provides a virtual, high speed solution for identifying individuals in environments like:

- College Classrooms for Attendance
- Office Entrances for security enhancement
- Restricted Lab Access

Problem Statement

Traditional methods of identification like ID cards and manual verification are outdated as they are slow and have errors and there might be human interferences which can cause disparity. In this project we aim to develop a face detection and recognition system using Deep Learning which will provide a fast, accurate and contactless ways to identify individuals in real time.

Objectives

- To develop a real time face detection system using the Haar Cascade classifier.
- To implement a face recognition module using the Local Binary Patterns Histograms (LBPH) algorithm.
- To create a system to store and train on a custom dataset of authorized individuals.
- To provide a contactless and automated alternative for attendance or security system.

Methodology

Step 1: Data Collection (Creating the Register)

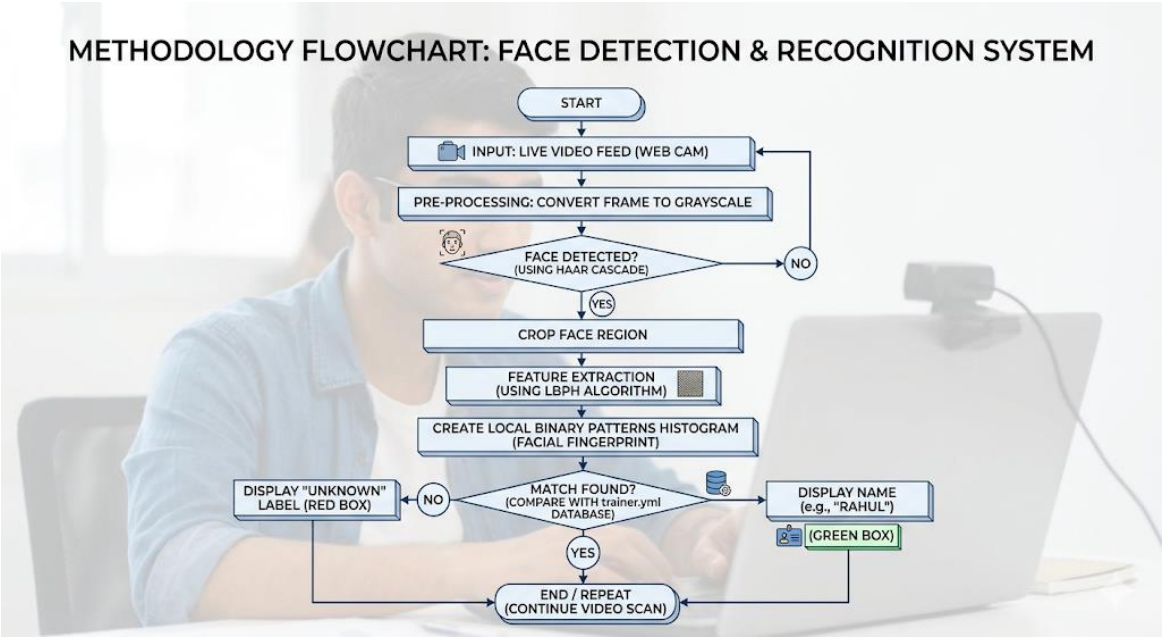
- The camera captures about 30–50 photos of a person.
- The images are converted to Grayscale (Black and White) to make the computer work faster.
- Each person is given a unique ID number (e.g., Student 1, Student 2).

Step 2: Training (Learning the Faces)

- The system uses the LBPH algorithm to "study" these photos.
- It creates a mathematical "map" or fingerprint of each face based on skin texture and features.
- This map is saved into a small file called trainer.yml.

Step 3: Real-Time Recognition (Testing)

- Detection: Using Haar Cascade, the camera looks for any face in the room.
- Matching: Once it finds a face, it compares it to the maps in the trainer.yml file.
- Result: If a match is found, it displays the Name on the screen. If not, it says unknown.



Software Tools

- Python: The core programming language. It is chosen for its simplicity and the powerful OpenCV library.
- OpenCV (Open Source Computer Vision): The main library used to process images, detect faces, and run the recognition algorithms.
- NumPy: A library used to handle image data. Since a digital image is just a grid of numbers, NumPy helps the computer do the math very quickly.
- VS Code / PyCharm / IDLE: The Integrated Development Environment (IDE) used to write and run the Python scripts.

Hardware Requirements

- Webcam: To capture the live video feed.
- Standard PC/Laptop: No heavy Graphics Card (GPU) is required for this specific project, making it very accessible.

Techniques

These are the two main "engines" that make your project work:

1 .Haar Cascade Classifier (The Detection Technique)

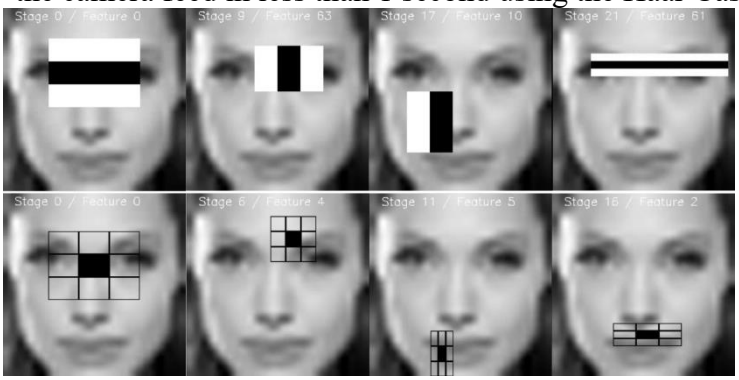
- Purpose: To find "Where" the face is.
- Logic: It uses Paul Viola and Michael Jones's method. It looks for specific facial features like the dark horizontal line of the eyes or the vertical bright line of the nose bridge.
- Key Advantage: It is extremely fast because it uses "Integral Images" to calculate these features instantly.

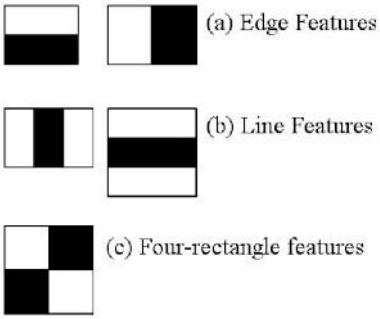
2. LBPH - Local Binary Patterns Histograms (The Recognition Technique)

- Purpose: To identify who the person is.
- Logic: 1. It breaks the face into small cells. 2. It compares each pixel with its neighbours to create a binary pattern (1s and 0s). 3. It creates a Histogram (a graph) of these patterns.
- Key Advantage: It is "robust" against lighting changes. Even if the room is a bit dark, the texture pattern of the skin remains the same for the computer.

Expected Outcomes

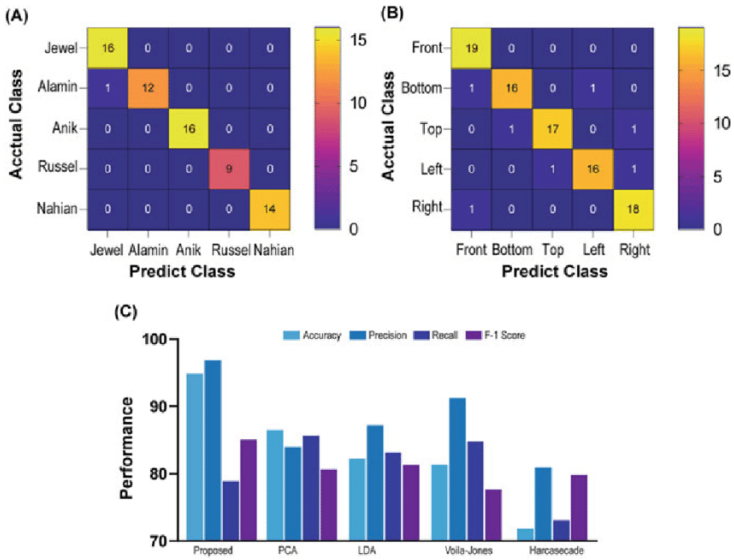
- **Real-Time Face Detection:** The system should successfully identify a human face in the camera feed in less than 1 second using the Haar Cascade classifier.





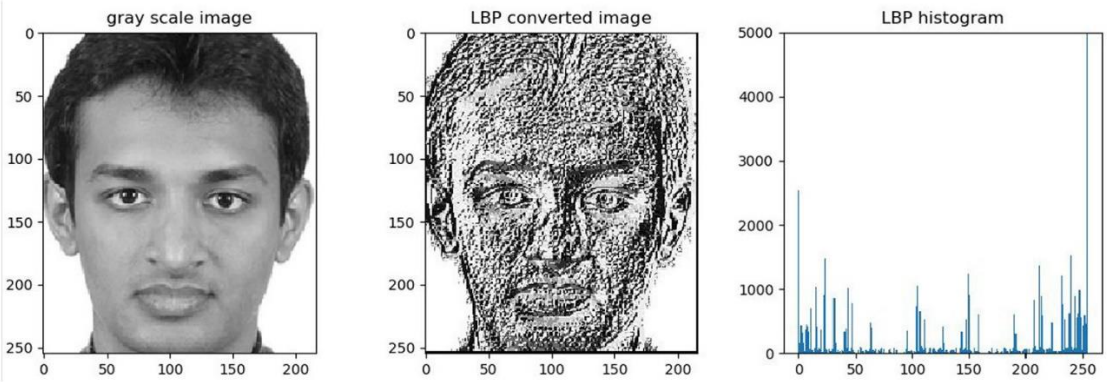
- **High Recognition Accuracy:** The system is most likely to correctly identify registered individuals (students/staff) with an accuracy of 90% or higher under normal indoor lighting conditions.

- **Successful Unknown Labelling:** If a person who is not in the database stands in front of the camera, the system should correctly label them as Unknown instead of guessing a wrong name.



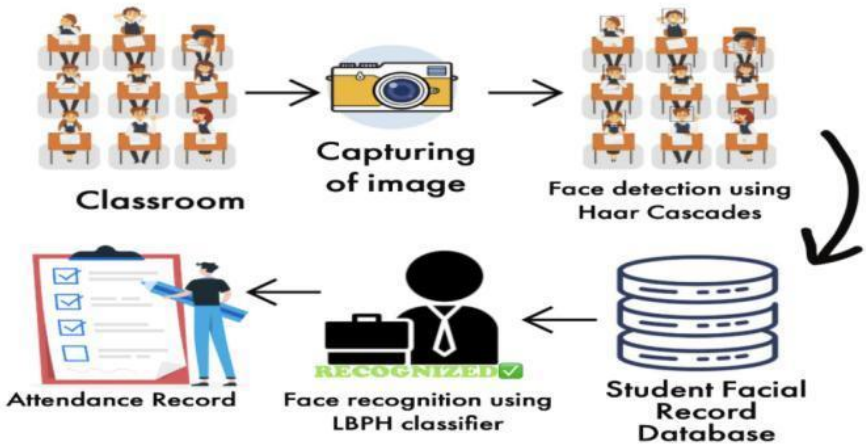
- **Low Computational Cost:** The project should run smoothly on a standard laptop (CPU-based) without needing an expensive Graphics Card (GPU), maintaining a steady frame rate (FPS).

- **Confidence Score Display:** For every recognized face, the system will provide a "Confidence Score" to show how certain the computer is about the person's identity.



Applications

- **Smart Attendance:** Automatically marking students Present in a digital register as they enter the classroom.
- **Secure Access Control:** Unlocking laboratory or office doors only for authorized staff without using keys or ID cards.
- **Contactless Check-in:** Identifying people at hospital receptions or airports without any physical contact, which is more hygienic.
- **Personalized Greeting:** Systems like Smart Mirrors or PCs that recognize the owner and display their personal schedule or "Hello [Name]."
- **VIP Identification:** Helping retail stores or hotels recognize frequent customers to provide them with better, faster service.



Conclusion

This project successfully demonstrates a low cost, high speed way to identify people in real time. By using Python and OpenCV, we have created a Touchless security solution that is much faster and more reliable than traditional ID cards or manual registers.

Future Work

- **Mobile Integration:** Sending a notification to a teacher's phone when a student is recognized.
- **Liveness Detection:** Adding a blink check to make sure the person is real and not in a photo or some other person with similar faces.
- **Cloud Database:** Storing all attendance records on a website so they can be accessed from anywhere.

References

- **Images**

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